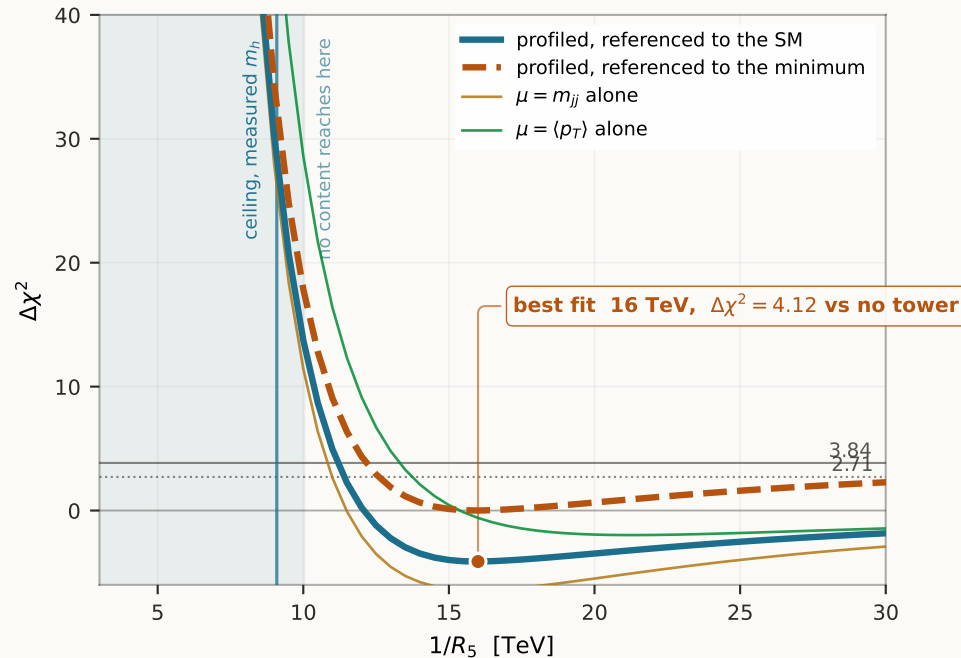
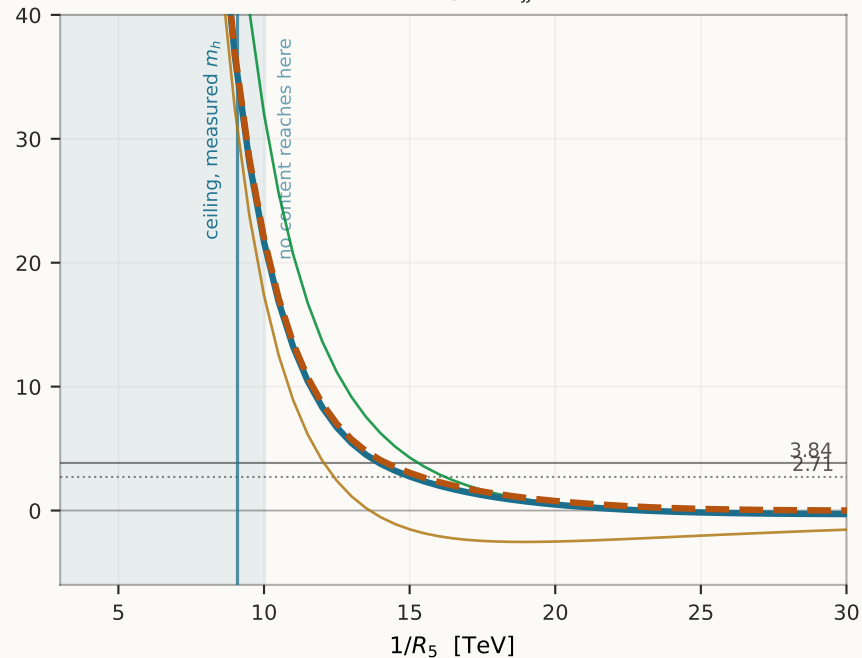


The likelihood, and the per-rung ceiling envelope laid on it

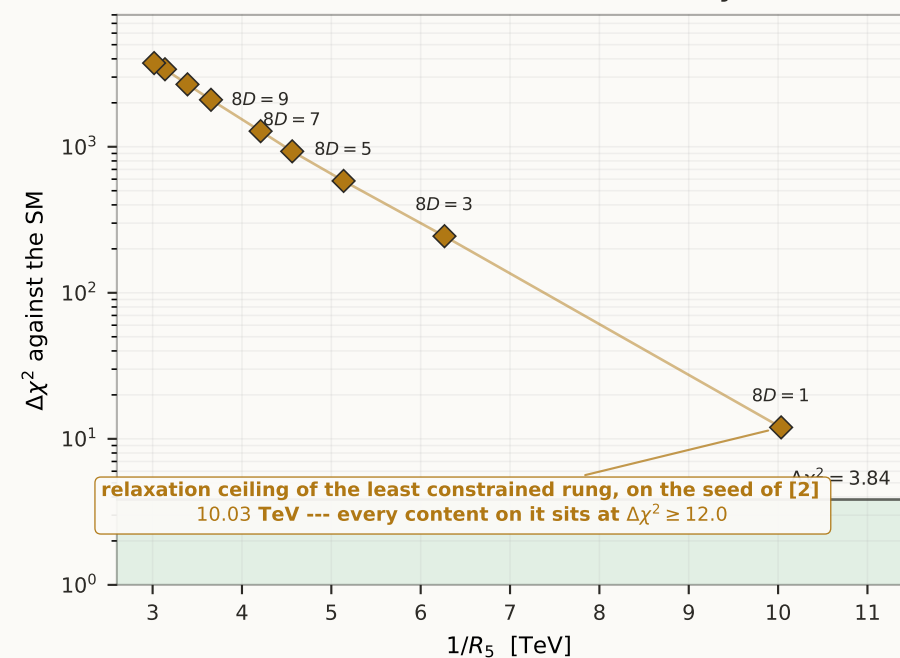
All seven mass bins



CMS's own range, $M_{jj} > 3.6$ TeV



Where the allowed models actually fall



Left: the data prefer a finite tower over none by $\Delta\chi^2 = 4.12$. Middle: drop the 3.0--3.6 TeV bin --- the one carrying CMS's own 2.0σ local excess --- and the preference collapses to 0.29, so it is theirs and not ours. Right: the question this paper can ask and a continuous lower limit cannot --- the relaxation ceiling of each reachable rung, against the same likelihood. Contents sit at or below theirs, so by monotonicity these are conservative lower bounds and not attained points.